SVKM'S NMIMS

MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT& ENGINEERING/

SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Program/s: B.Tech /MBA Tech/BTI

Year: II/IV Semester: III/VII

Stream/s : Data Science

Subject: Managing Uncertainty

Time: 3 hrs (10.00 a.m to 1.00 p.m)

Date: 22/2/25

No. of Pages:5

Marks: 100

Re Examination (2024-25) /Re-Examination (2022-23, 23-24)

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

1) Question No. _1___ is compulsory.
2) Out of remaining questions, attempt any __4___ questions.
3) **In all ___5_ questions to be attempted.**
4) All questions carry equal marks.
5) **Answer to each new question to be started on a fresh page.**
6) **Figures in brackets on the right hand side indicate full marks.**
7) **Assume Suitable data if necessary.**

<table>
<tr><td>Q1</td><td></td><td>Answer briefly:</td><td>[20]</td></tr>
<tr><td>CO-3 ;
SO-2 ;
BL-2</td><td>a.</td><td>Explain the objective of Tabulation. Explain the difference between classification and tabulation</td><td>[5]</td></tr>
<tr><td>CO-3 ;
SO-2 ;
BL-3</td><td>b.</td><td>The data below give the yearly profits (in thousands of rupees) of two companies A and B. Develop the data by means of a suitable bar diagram.
<table>
<tr><td>Year</td><td>Company A</td><td>Company B</td></tr>
<tr><td>1994-95</td><td>120</td><td>90</td></tr>
<tr><td>1995-96</td><td>135</td><td>95</td></tr>
<tr><td>1996-97</td><td>140</td><td>108</td></tr>
<tr><td>1997-98</td><td>160</td><td>120</td></tr>
<tr><td>1998-99</td><td>175</td><td>130</td></tr>
</table></td><td>[5]</td></tr>
</table>

<table>
<tr><td>CO-1 ;
SO-2 ;
BL-1</td><td>c.</td><td>An urn contains 8 white and 3 red balls. If two balls are drawn at random, find the probability that (i) both are white, (ii) both are red, (iii) one is of each color.</td><td>[5]</td></tr>
<tr><td>CO-2 ;
SO-2 ;
BL-2</td><td>d.</td><td>Explain Type I and Type II error</td><td>[5]</td></tr>
<tr><td>Q2.a
CO-3;
SO-2;
BL-5</td><td></td><td>Determine Karl-Pearson's coefficient of skewness for the given distribution:
<table>
<tr><td>X:</td><td>0-5</td><td>5-10</td><td>10-15</td><td>15-20</td><td>20-25</td><td>25-30</td><td>30-35</td><td>35-40</td></tr>
<tr><td>F:</td><td>2</td><td>5</td><td>7</td><td>13</td><td>21</td><td>15</td><td>8</td><td>3</td></tr>
</table></td><td>[10]</td></tr>
<tr><td>Q2.b
CO-3;
SO-2;
BL-5</td><td></td><td>For the following frequency distribution, determine (a) The median class. (b) The number of the item that represents the median. (c) The width of the equal steps in the median class. (d) The estimated value of the median for these data. (e) Estimate the value of Median with the formula.
<table>
<tr><th>Class in Dollars</th><th>Frequency</th></tr>
<tr><td>0–49.99</td><td>78</td></tr>
<tr><td>50.00– 99.99</td><td>123</td></tr>
<tr><td>100.00–149.99</td><td>187</td></tr>
<tr><td>150.00–199.99</td><td>82</td></tr>
<tr><td>200.00–249.99</td><td>51</td></tr>
<tr><td>250.00–299.99</td><td>47</td></tr>
<tr><td>300.00–349.99</td><td>13</td></tr>
<tr><td>350.00–399.99</td><td>9</td></tr>
<tr><td>400.00–449.99</td><td>6</td></tr>
<tr><td>450.00–499.99</td><td>2</td></tr>
<tr><td></td><td>600</td></tr>
</table></td><td>[10]</td></tr>
<tr><td>Q3.a
CO-1;
SO-2;
BL-6</td><td></td><td>We have data from 100 economists in academics, private sector, and government, concerning their opinions whether the economy would be stable, expand or contract in the near future. However, a part of the information was lost. Based on remaining data, you are required to create a probability table.
From the given table find : (i) P(A) ; (ii) P(E) ; (iii) P(A ∩ E) ;
(iv) P(G ∩ C) ; (v) P (E | G) ; (vi) P (G | E).</td><td>[10]</td></tr>
</table>

		<table><tr><th rowspan="2">Economists</th><th colspan="3">Economy</th><th rowspan="2">Total</th></tr><tr><th>Stable (S)</th><th>Expanding (E)</th><th>Contracting (C)</th></tr><tr><td>Academics (A)</td><td>25</td><td></td><td>20</td><td></td></tr><tr><td>Private Sector (R)</td><td></td><td>7</td><td></td><td>22</td></tr><tr><td>Government (G)</td><td>5</td><td>8</td><td></td><td>13</td></tr><tr><td></td><td>40</td><td></td><td></td><td></td></tr></table>	
Q3.b CO-1; SO-2; BL-5		The results of an investigation by an expert on a fire accident in a skyscraper are summarized below : (i) Prob. (there could have been short circuit) = 0·8 (ii) Prob. (LPG cylinder explosion) = 0·2 (iii) Chance of fire accident is 30% given a short circuit and 95% given an LPG explosion. Based on these, what do you think is the most probable cause of fire? Statistically justify your answer.	[10]
Q4.a CO-2; SO-2; BL-5		The random variable X has binomial distribution B 15, 0.35 ( ). Determine each of the following. a) P (X = 8). b) P (X < 7). c) P (X > 9). d) P (5< X< 10).	[10]
Q4.b CO-2; SO-2; BL-5		A continuous function of a random variable X has the p.d.f $f(x) = kx^2e^{-x}$, x>=0. Evaluate k, mean and variance.	[10]
Q5.a CO-2; SO-2; BL-3		It is given that 2% of the test tubes are defective. Apply Poisson Distribution to find the probability that a packet of 100 test tubes contains: (1) No defective test tubes (2) One defective test tubes (3) Two or more defective test tubes (4) Two or less defective test tubes.	[10]
Q5.b CO-2; SO-2; BL-		Suppose that in an Olympics game Archer shoots a target in an independent fashion. If the probability that the target is hit in any shot is 0.7. Determine the (i) Probability that the target would hit on the 10th attempt?(ii) probability that it taken him less than 4 shots? (iii) Probability that it taken him an even number of shots? (iv) Average number of shots needed to hit the target.	[10]
Q6.a		The joint density function of A and B is given by P(a, b) = k(2a+b); a = 0,1,2; b = 0,1,2	[10]

<table>
<tr><td>CO-1;
SO-2;
BL-4</td><td></td><td>Distinguish (i) value of k (ii) Marginal distribution of A and B (iii) are A and B independent? (iv) find the conditional distribution of B for A = a.</td><td></td></tr>
<tr><td>Q6.b
CO-3;
SO-2;
BL-4</td><td></td><td>In a population, the average IQ is 100. A team of scientists wants to test a new medication to see if it has either a positive or negative effect on intelligence or no effect at all. A sample of 30 participants who has taken the medication has a mean of 140 with a standard deviation of 20. Did the medication affect intelligence? Alpha = 0.05</td><td>[10]</td></tr>
<tr><td>Q7.a
CO-3;
SO-2;
BL-2</td><td></td><td>State Bayes theorem. Explain in detail with proof</td><td>[10]</td></tr>
<tr><td>Q7.b
CO-1;
SO-2;
BL-4</td><td></td><td>Two independent random samples of size 4 and 5 are selected from each of the normal populations
<table>
<tr><td>Population 1</td><td>12</td><td>3</td><td>8</td><td>5</td><td></td></tr>
<tr><td>Population 2</td><td>14</td><td>7</td><td>7</td><td>9</td><td>6</td></tr>
</table>
(1) calculate S2 the pooled estimator of variance
(2) find the 95% confidence interval for $\mu 1 - \mu 2$, the difference between the two population
(3) Test H0: $\mu 1 - \mu 2 = 0$ against H1: $\mu 1 - \mu 2 \neq 0$ for alpha = 0.05. state your conclusion</td><td>[10]</td></tr>
</table>

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										